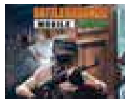
75 years of India's Independence



BGMI Ban
BGMI ban in India has reportedly resulted in the game being removed from app stores.
https://dgit.in/aug22-37

AMD Ryzen 7 5700X

Earlier would have been better

₹26,999

With the next family of processors from AMD on the horizon, it was an odd thing for them to release the Ryzen 7 5700X in April of 2022. Most of the Ryzen 5000 series processors have been out for a good year and a half, and Intel has only recently put out their 12th Gen Alder Lake processors with SKUs that are actually competing and even beating AMD Ryzen 5000 SKUs in some segments. And Intel having more SKUs at price points which had no competing AMD processors is the reason why we're seeing the Ryzen 7 5700X this late into the cycle. So is it a good competitor or is this an example of too little and too late?

SPECIFICATIONS

April 2022 saw several AMD Zen 3 processors being launched. There were several 4000 series SKUs in the lower price points which were completely ignored by AMD during the initial Ryzen 5000 launch. And since Intel has several SKUs in that segment, AMD finally released several SKUs to compete with them. The AMD Ryzen 7 5700X, also part of the April 2022 launch cycle, is a Vermeer chip on the inside. It's basically a scaled down Ryzen 7 5800X with lower clock speeds but the same set of cores. The base clock speed is 3.4 GHz and the boost clock speed is 4.6 GHz. It would seem that if thermals aren't a problem, then the performance of the two SKUs might end up being very similar. One of the biggest changes would be the TDP which is 65 Watts for the 5700X. That's 40 Watts less than the 5800X.

Pricing on the 5700X is a tad bit lesser than the 5800X but what remains to be seen is if the price difference makes it a worthwhile option to consider. While the 5800X lies on the higher end of the performance graph, there's also the 5700G for folks to consider. The 5700G is a Cezanne chip with an inbuilt graphics core and has a configurable TDP range of 45-65 Watts. It sports higher clock speeds as well but the key difference being the PCIe lanes emerging from the CPU is Gen 3 on the 5700G vs Gen 4 on the 5700X. So there are plenty of reasons for the two SKUs to exist side-by-side.

PERFORMANCE

Onto the big question - Is it worth the money? We start off with Cinebench R23 where the 5700X scores 13,128 while the 5800X scored 15,147. That's a 15 per cent difference in a synthetic benchmark. Switching to Blender gives us a 13 per cent difference. Being an unlocked processor, we put the screws on the 5700X and managed to get it up to 4.4 GHz and at those clocks, the 5700X managed to nearly equal the 5800X in Blender.

In our productivity benchmarks, the Ryzen 7 5700X was very similar to the 5800X without any overclocking. Precision Boost Overdrive does a good job on its own. Switching to content creation benchmarks painted a slightly different picture since the workloads would run for slightly longer periods of time and the TDP limits were kicking in. Nevertheless, the performance difference wasn't that much at about 7 per cent. Overclocking again helped the 5700X equal the 5800X in these benchmarks.

Lastly, we moved to gaming. There's literally negligible difference between the 5700X and the 5800X in gaming. We're talking about 1-1.5 per cent difference in FPS values. And that's without overclocking. In fact, overclocking didn't help here at all. So we're looking at a processor that can equal the 5800X with a significant overclock in most benchmarks but requires none of it when you're gaming. If you're getting a really good cooler, then PBO should do most of the work.

76

PERFORMANCE..........77
VALUE FOR MONEY...75

VERDICT

The 5700X costs about ₹24,750 and the 5800X costs ₹27,800. That's a 12 per cent price difference at the time of writing this review. And the performance difference between the two SKUs is also within the same range. So in a world where there aren't any overclocking technologies such as PBO, the two processors can live comfortably next to each other. But all things considered, the 5700X comes across as a better value for money offering for gaming and most day-to-day productivity tasks. Only sustained workloads bring out the difference between the two SKUs making the 5700X a better 8-Core/16-Thread recommendation. Now you can see why the 5700X wasn't launched when the rest of the Zen 3 processors hit the market.

–Mithun Mohandas

SPECIFICATIONS
CORES: 8. | THREADS: 16 | GPU CORES: NA | BASE CLOCK: 3.4 GHz | BOOST CLOCK: 4.6 GHz | PROCESS NODE: 7nm | L3 CACHE: 32 MB | L2 CACHE: 512 KB per core | TDP: 65 W.

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